

AIStudio — File Guide

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How to use this guide

This is a **functional/relational** file guide — answers "what does each file do, what does it consume, what does it produce, what depends on it" rather than just "where does it live."

§1a File Versioning Conventions

AIStudio uses two versioning patterns depending on file type. Both are mandatory for any file that is deployed or tracked.

Shell scripts (`.sh`)

Two lines, both required, **both must match**:

```
# Version: X.Y.Z
VERSION="X.Y.Z"
```

Python files (`.py`)

Header comment block at the absolute top of the file, before all imports:

```
# Version: X.Y.Z
# Changelog: X.Y.Z — <ticket> description of change.
#           Continuation lines indented 12 spaces (aligned with description start).
# Changelog: X.Y.Z-1 — previous change (newest first).
```

Rules: - `# Version:` must match the most recent `# Changelog:` entry — single source of truth. - Entries are prepended (newest first) — never append at the bottom. - One entry per version bump. Two tickets in one bump is fine. - The header block appears before `from` `__future__` `import annotations` and all imports.

Class C files — version inside file, not in filename

Per §1, Class C (code/SDLC) files carry no date in the filename. The `# Version:` header is the only version signal.

§2 The runtime layer — `src/` + `front_end/` + `data/`

What runs to answer a query.

`src/local_llm_bot/app/` — Python application

File	Function	Reads	Produces
<code>api.py</code>	FastAPI HTTP layer	Corpus metadata, query payload, system prompt	JSON response with citations
<code>rag_core.py</code>	RAG pipeline orchestration	Query, corpus index	Embedded query → retrieved chunks → reranked → LLM context
<code>ingest/pipeline.py</code>	Document ingestion (per-file chunked/ embedded/ upserted)	Files in <code>data/corpora/<name>/</code> <code>uploads/</code> , corpus metadata seed	<code>index.jsonl</code> , <code>manifest.jsonl</code> , Qdrant chunks
<code>ingest/loaders.py</code>	File-type-specific loaders	PDF/DOCX/HTML/ MD files	Plain text + page metadata
<code>ingest/chunking.py</code>	Text chunking with overlap	Plain text	Chunks with stable IDs
<code>vectorstore/qdrant_store.py</code>	Vector store interface	Embeddings, metadata	Qdrant collections, similarity search
<code>ollama_client.py</code>			

File	Function	Reads	Produces
	Ollama HTTP wrapper	Model name, prompt	Embedding vectors, LLM completions
<code>config.py</code>	Environment-driven config	OS env vars	Settings dict

`front_end/rag_studio.html` — UI

Single self-contained HTML file. Opens directly in browser, no server. Communicates with FastAPI backend at `localhost:8000`.

Provides: corpus selector, chat with citations, corpus CRUD (new/delete/rename), file upload + per-file inspect, metadata edit (description, `search_guidance`), help modal.

`data/corpora/<name>/` — Document corpora

Each corpus is a self-contained folder:

File	What it is	Who writes it
<code>uploads/</code>	Source documents (PDF, DOCX, etc.)	User via UI Add Files
<code><name>_corpus_metadata.yaml</code>	Description, <code>search_guidance</code> , <code>sources[]</code> , default query settings, runtime stats	Seed at create-time, updated at ingest; UI Edit Corpus
<code><name>_full_scope.yaml</code>	Corpus membership — the firm/entity universe (the superset). At the corpus root.	Downloader / entity-KB tools (<i>bundled financial corpora</i>)
<code>scopes/</code> <code><name>_<stem>_scope.yaml</code>	Named subsets of the membership (e.g. <code>lang_en</code> , <code>big_banks</code>) — read-only selectors for download, bench, selective ingest	Hand-authored (<i>optional</i>)
<code>index.jsonl</code> , <code>manifest.jsonl</code>	Internal ingest bookkeeping (per-chunk / per-file records). Not user-edited; internal, may be retired in a future	Ingest pipeline

File	What it is	Who writes it
	cleanup — corpus config now lives in <code><name>_corpus_metadata.yaml</code> .	
<code>trash/</code>	Soft-deleted files (recoverable)	UI delete

Note the split: a corpus's **defining inputs** — documents, metadata, and (for the financial corpora) scope + entity KB — live under `data/corpora/<name>/`. Benchmark **question sets** live separately under `benchmarks/<name>/` (next section).

Demo corpus ships with the repo. **Help corpus** is built from `docs/` + key root markdown. **SEC 10-K corpus** is downloaded from SEC EDGAR via `ais_download_sec_10k` and ingested through the UI (Upload button), like any corpus. **User corpora** are created via the UI New button — private, gitignored, no special setup required.

`benchmarks/<corpus>/` — Benchmark questions and reports

`benchmarks/<corpus>/` holds the bench **question sets** (inputs) and the **reports** they produce (outputs) — kept separate from the corpus-defining files under `data/corpora/<corpus>/`. Each corpus that has a questions file can be tested with `ais_bench`:

```
benchmarks/
  demo/
    demo_questions.yaml      ← pre-built questions for the demo corpus
    reports/                 ← timestamped benchmark reports (.md, .json, .pdf)
  sec_10k/
    sec_10k_questions.yaml   ← default question set
    sec_10k_<stem>_questions.yaml ← named alternates (e.g. sec_10k_June_2026_questions.yaml)
    reports/
  <your-corpus>/
    <your-corpus>_questions.yaml ← you write this (see format below)
    reports/
```

Shipped evidence suite — `benchmarks/docs/` holds the curated, user-facing benchmark suite: the four audited canonical reports (`BENCH - Canonical Run A-D - ...`, `.md` + `.pdf`), the suite synthesis (`BENCH - Canonical Suite - README and Synthesis`), and `benchmarks/docs/reports/` (the raw `ais_bench` reports duplicated as provenance). This is what the top README points to as the evaluation evidence; reproduce it with `ais_bench --canonical`, which runs each entry on its pinned model. (`ais_bench --batch` runs the same question sets against whatever your machine can hold — useful for characterising your own hardware, but it does not reproduce the published figures.)

The default set is `<corpus>_questions.yaml` ; a named alternate is selected by stem (`ais_bench --questions June_2026 → sec_10k_June_2026_questions.yaml`). Reports are written per run into `reports/` under a standard timestamped name — `.md` to read, `.json` for tooling, `.pdf` to share.

Questions file format — create `benchmarks/<name>/<name>_questions.yaml` :

```
- topic: Topic Name
  questions:
    - id: unique_snake_case_id
      question: The exact question text sent to AISTudio.
      keywords: [term1, term2, term3] # all must appear in the answer to pass
      notes: What a correct answer looks like — which document, what content.
```

What `ais_bench` checks — three conditions, all must pass: 1. All `keywords` appear in the answer (case-insensitive) 2. The answer includes at least one citation 3. The model doesn't hedge with "no information available" or similar phrases

Tips for good keywords: 2–5 distinctive terms that prove the model retrieved the right content. Use specific concepts, entity names, or regulatory terms — not generic words.

For full details and advanced options, see `TUTORIAL.md §3.4` and `HOWTO.md` .

`prompts/system.txt` — System prompt

Loaded by `api.py` at query time. Per-corpus `search_guidance` from `<name>_corpus_metadata.yaml` is appended dynamically.

§3 The user-facing command surface

All commands available after running `./ais_install` from repo root.

Command	What it does	When you use it	Reads / Produces
<code>ais_start</code>	Starts FastAPI + Qdrant + Ollama; opens UI	Beginning of every session	— / running services
<code>ais_stop</code>		End of every session	— / —

Command	What it does	When you use it	Reads / Produces
	Stops all services cleanly		
<code>ais_log</code>	Tails the live backend log	Debugging; watching queries	service logs / stdout
<code>ais_bench</code>	Runs benchmark suite against a corpus	Validating retrieval quality	<code>benchmarks/<corpus>/<corpus>_questions.yaml</code> / benchmark report
<code>ais_download_sec_10k</code>	Downloads SEC 10-K filings from EDGAR	Setting up SEC corpus (optional)	EDGAR / <code>data/corpora/sec_10k/uploads/</code>
<code>ais_help</code>	Shows command reference	Forgot a command	<code>ais_command_help.txt</code> / stdout
<code>ais_download_esef</code>	Downloads ESEF iXBRL annual reports from filings.xbrl.org	Setting up ESEF corpus	<code>esef_banks_full_scope.yaml</code> / <code>data/corpora/esef_banks/uploads/</code>
<code>ais_ingest_esef</code>	Ingests ESEF filings into AIStudio	After downloading ESEF	<code>uploads/</code> / Qdrant
<code>ais_import_entity_kb</code>	Builds entity KB for a corpus (GLEIF/ Wikidata identity + aliases)	After ingest; after adding a firm	<code><corpus>_full_scope.yaml</code> / <code>gleif_<corpus>_full_entities.yaml</code>
<code>ais_import_glossary_kb</code>	Builds glossary KB (BIS Basel	After ingest; corpus-wide	static seed / <code>bis_basel_any_corpus_full_glossary.yaml</code>

Command	What it does	When you use it	Reads / Produces
	acronym expansion for BM25)		
<code>ais_install</code>	Installs/ updates user commands	Fresh install; new command	<code>ais_user_commands_manifest.yaml</code> / shell aliases in <code>~/ .zshrc</code>
<code>ais_create_shortcut</code>	Creates AIStudio Dock/ Desktop icon (macOS)	Optional post-install	/ <code>~/Desktop/AIStudio.app</code>

§9 The repo root layout

User-visible (public, ships with repo)

File / Dir	What it does	Relationship
<code>README.md</code>	Product overview	Entry point for new users
<code>QUICKSTART.md</code>	Install + first-run guide	Read after README
<code>HOWTO.md</code>	Day-to-day usage recipes	Read for "how do I X"
<code>TUTORIAL.md</code>	Guided walkthroughs	Read after QUICKSTART
<code>CONTRIBUTING.md</code>	Contribution guide	Read for repo participation
<code>LICENSE</code>	License terms	Repo convention
<code>Makefile</code>	<code>make</code> <code>install</code> , <code>make test-</code>	Build orchestration

File / Dir	What it does	Relationship
	<code>unit</code> , <code>make test</code> , <code>make lint</code>	
<code>pyproject.toml</code>	Python project config (ruff, pytest)	Tool config
<code>requirements.txt</code>	Python dependencies	<code>pip install -r</code>
<code>ais_install</code>	User command installer	Reads <code>ais_user_commands_manifest.yaml</code>
<code>ais_user_commands_manifest.yaml</code>	Auto-generated user command list	Produced by install
<code>ais_*.sh</code> (user)	User commands (see §3)	Aliased by <code>ais_install</code>
<code>prompts/system.txt</code>	LLM system prompt	Loaded by <code>api.py</code>
<code>front_end/rag_studio.html</code>	UI	Opened in browser
<code>src/</code>	Application source (see §2)	Imported by <code>api.py</code>
<code>tests/</code>	Test suite	Run by <code>make test</code>
<code>docs/</code>	User documentation + help corpus sources	Ingested into help corpus
<code>benchmarks/</code>	Benchmark harness + question files + reports	Run by <code>ais_bench</code>

File / Dir	What it does	Relationship
<code>data/</code>	Corpus data (demo + help tracked; user corpora gitignored)	Read by <code>rag_core.py</code>

§12 The file-to-action map

Quick lookup: who reads what, who writes what, what feeds into what.

File pattern	Reads	Writes	Feeds into
<code><corpus>_corpus_metadata.yaml</code>	<code>api.py</code> (system prompt), UI Edit	Ingest pipeline, UI Edit	LLM routing, search_guidance display
<code>prompts/system.txt</code>	<code>api.py</code>	Manual edit	LLM system instructions
<code>data/corpora/<name>/uploads/*</code>	Ingest pipeline	UI Add Files	Corpus chunks → Qdrant
<code>data/corpora/<name>/index.jsonl</code>	(<i>internal — see note</i>)	Ingest pipeline	Per-chunk ingest record
<code>benchmarks/<corpus>/<corpus>_questions.yaml</code>	<code>ais_bench</code>	User-authored or auto-generated	Benchmark pass/fail evaluation

§12a What files define a corpus — and where they live

This is the *where* for corpora; the *how* (what builds them) is in `CODEBASE_GUIDE.md` → "How a corpus is defined, built, and used". Every corpus — the built-in ones and any you create — lives under `data/corpora/<name>/`.

What the app reads

File / dir	Purpose
<code>data/corpora/<name>/uploads/</code>	the actual documents (what gets searched)
<code>data/corpora/<name>/<name>_corpus_metadata.yaml</code>	the corpus's description, search guidance, and default query settings; read every time you ask a question
<code>data/corpora/<name>/<name>_full_scope.yaml</code>	(bundled <code>sec_10k</code> / <code>esef_banks</code> only) the corpus membership — which firms/entities belong to it
<code>data/corpora/<name>/scopes/<name>_<stem>_scope.yaml</code>	(optional) named subsets of the membership (e.g. <code>lang_en</code>) — selectors for download, benchmarking, and selective ingest
<code>benchmarks/<name>/<name>_questions.yaml</code>	(optional) benchmark question sets for the corpus — note these live under <code>benchmarks/</code> , not with the corpus
<code>data/knowledge_sources/gleif/<name>_entities.yaml</code>	(bundled <code>sec_10k</code> / <code>esef_banks</code> only) company-identity (GLEIF) lookups used to keep firms apart in cross-company questions — see Tutorial Annex 1

So *what files define a corpus*? At minimum: the documents in `uploads/`, and the `<name>_corpus_metadata.yaml` that describes it. Membership scope, subset scopes, questions, and the GLEIF identity file are enrichment used by the bundled financial corpora.

Where things live — the split. A corpus's *defining* files sit together under `data/corpora/<name>/`: the documents, the metadata, and (for the financial corpora) the scope and entity KB. Benchmark *question sets* sit apart under `benchmarks/<name>/`, alongside the `reports/` they generate. Rule of thumb: **what defines the corpus** → `data/corpora/`; **how you test it** → `benchmarks/`.

Where YOUR files go

When you create your own corpus, everything lives under `data/corpora/<your-corpus>/`:

- `uploads/` — your files. Created and filled by the **Upload** button in the UI; you don't make this folder by hand.
- `<your-corpus>_corpus_metadata.yaml` — created by the **Edit Corpus** modal when you give the corpus a description, search guidance, and default settings.

- `benchmarks/<your-corpus>/<your-corpus>_questions.yaml` — only if you choose to add benchmark questions.

To query your own portfolio — your filings, reports, decks, or PDFs — create a corpus in **Settings** → **New Corpus**, then drop your files in via **Upload**. The corpus is defined by what's in its `uploads/` folder plus the description you give it. That's everything you need.

§15 Version history

Version	Date	Changes
Beta	2026-06-24	Corpus-data consistency sweep (§71, §85, §12a): added membership scope (<code><name>_full_scope.yaml</code>) + subset scopes (<code>scopes/</code>) to the corpus tables; stated the defining-inputs (<code>data/corpora/</code>) vs question-sets (<code>benchmarks/</code>) split explicitly; documented named question alternates by stem; softened <code>index.jsonl</code> / <code>manifest.jsonl</code> to internal-may-be-retired.
Beta	2026-06-15	Trimmed to the AIStudio file, command, and corpus reference; SEC ingest is UI-only.
1.2.0	2026-06-08	Added §12a "What files define a corpus — and where they live" (corpus definition + where user portfolio files go).
1.1.0	2026-05-25	Added user-created corpus type. Added benchmarks/ subsection with questions file format and pass/fail explanation. Added benchmarks row to §12.
1.0.0	2026-05-20	Initial version.

